

The cone extension of a convex set

Kent Andersen

A convex set in $\mathbb{R}^n$ can be viewed as a subset of a convex cone in $\mathbb{R}^{n+1}$, where the convex set is obtained from the convex cone by fixing the last coordinate to a constant. For instance, the unit sphere $\{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}$ can be viewed as the subset of the convex cone $\{x \in \mathbb{R}^{n+1} : \|x\|_2 \leq x_{n+1}\}$ obtained by fixing $x_{n+1} = 1$.

The cone extension of a convex set $K \subseteq \mathbb{R}^n$ is a convex cone $K^{\text{ext}} \subseteq \mathbb{R}^n \times \mathbb{R}_+$, where an extra non-negative variable is added:

$$K^{\text{ext}} := \text{cone}\left(\left\{\begin{pmatrix} x \\ 1 \end{pmatrix} : x \in K\right\}\right) + \text{cone}\left(\left\{\begin{pmatrix} x \\ 0 \end{pmatrix} : x \in \text{rec}(K)\right\}\right)$$

and one has $\begin{pmatrix} x \\ 1 \end{pmatrix} \in K^{\text{ext}} \iff x \in K$ and $\begin{pmatrix} x \\ 0 \end{pmatrix} \in K^{\text{ext}} \iff x \in \text{rec}(K)$. Every halfline of K^{ext} with positive last coordinate is obtained by scaling a point in K , and every halfline of K^{ext} with zero as last coordinate is obtained by scaling a recession direction of K .

A key example is a polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$. The recession cone of P is given by $\text{rec}(P) = \{x \in \mathbb{R}^n : Ax \leq 0_m\}$ (Theorem 1 in note 2.(C)), and the cone extension of P is denoted $C_p(A, b)$ and given by:

$$C_p(A, b) = \left\{\begin{pmatrix} x \\ \lambda \end{pmatrix} \in \mathbb{R}^n \times \mathbb{R}_+ : Ax - \lambda \cdot b \leq 0_m\right\}.$$

A second key example is a finitely generated set $Q = \text{conv}(V) + \text{cone}(E)$. The recession cone of Q is given by $\text{rec}(Q) = \text{cone}(E)$ (Theorem 2 in note 2.(C)), and the cone extension of Q is denoted $C_f(V, E)$ and given by:

$$C_f(V, E) = \text{cone}\left(\left\{\begin{pmatrix} v \\ 1 \end{pmatrix} : v \in V\right\}\right) + \text{cone}\left(\left\{\begin{pmatrix} e \\ 0 \end{pmatrix} : e \in E\right\}\right).$$

In the remainder we connect properties of a general convex set $K \subseteq \mathbb{R}^n$ to the properties of its cone extension K^{ext} .

1 Pointedness of the cone extension

We first prove that K^{ext} is pointed iff the recession cone $\text{rec}(K)$ is pointed.

Theorem 1 (*Pointedness of K^{ext} versus pointedness of $\text{rec}(K)$*)

One has $\text{rec}(K)$ is pointed $\iff K^{\text{ext}}$ is pointed. $\blacksquare$

Proof: Observe that, since the last coordinate of points in K^{ext} is non-negative, there exists $\binom{d}{\mu}$ non-zero such that $-\binom{d}{\mu}, \binom{d}{\mu} \in K^{\text{ext}} \iff$ there exists d non-zero such that $-\binom{d}{0}, \binom{d}{0} \in K^{\text{ext}}$.

It follows that K^{ext} is *not* pointed $\iff K^{\text{ext}}$ contains a line $\iff$ there exists $\binom{d}{\mu}$ non-zero such that $-\binom{d}{\mu}, \binom{d}{\mu} \in K^{\text{ext}}$ holds $\iff$ there exists d non-zero such that $-\binom{d}{0}, \binom{d}{0} \in K^{\text{ext}} \iff$ there exists d non-zero such that $-d, d \in \text{rec}(K) \iff \text{rec}(K)$ contains a line $\iff \text{rec}(K)$ is *not* pointed. $\blacksquare$

2 Extreme directions of the cone extension

Extreme directions are important directions in a convex cone. A key property of the cone extension K^{ext} of K is that the extreme directions of K^{ext} with positive last coordinate correspond one-to-one to the extreme points of K :

Theorem 2 (*Extreme points of K versus extreme directions of K^{ext}*)

A direction $\binom{x}{1}$ is extreme in $K^{\text{ext}} \iff x$ is an extreme point of K . $\blacksquare$

Proof: " $\Rightarrow$ ": For a contradiction suppose $\binom{x}{1}$ is extreme for K^{ext} , but x is *not* an extreme point of K . This implies one may write $x = \lambda \cdot x^1 + (1 - \lambda) \cdot x^2$, where $\lambda \in]0, 1[$ and $x^1, x^2 \in K \setminus \{x\}$ (Theorem 2 in note 2.(A)). However, then $\binom{x}{1} = \lambda \cdot \binom{x^1}{1} + (1 - \lambda) \cdot \binom{x^2}{1}$, and neither of $\binom{x^1}{1}, \binom{x^2}{1} \in K^{\text{ext}}$ are equivalent to $\binom{x}{1}$. This contradicts that $\binom{x}{1}$ is extreme for K^{ext} .

" $\Leftarrow$ ": Now suppose, for a contradiction, that $\bar{x}$ is an extreme point of K , but $\binom{\bar{x}}{1}$ is *not* an extreme direction of K^{ext} . This implies $\binom{\bar{x}}{1}$ is a conic combination of: (1) directions $\binom{x}{1} \in K^{\text{ext}}$ with $x \in K$ non-equivalent to $\binom{\bar{x}}{1}$, and (2) directions $\binom{d}{0}$ with $d \in \text{rec}(K)$ non-equivalent to $\binom{\bar{x}}{1}$, i.e., one can write:

$$\binom{\bar{x}}{1} = \sum_{i \in I} \delta_i \cdot \binom{x^i}{1} + \binom{\bar{d}}{0} = \sum_{i \in I} \delta_i \cdot \binom{x^i + \bar{d}}{1} \quad (1)$$

where I is a finite index set, $\delta_i > 0$ and $x^i \in K \setminus \{\bar{x}\}$ for $i \in I$ and $\bar{d} \in \text{rec}(K)$. Hence $\bar{x}$ is a convex combination of the points $x^i + \bar{d}$ for $i \in I$. One must have $\bar{d} \neq 0$, since otherwise (1) gives $\bar{x}$ as a convex combination of the points $x^i \in K \setminus \{\bar{x}\}$ for $i \in I$, which contradicts that $\bar{x}$ is an extreme point of K .

Since $\bar{d}$ is a recession direction for K , one has $\{x^i + \alpha \cdot \bar{d} : \alpha \geq 0\} \subseteq K$ for all $i \in I$. Therefore, if $\bar{x} \in \{x^i + \alpha \cdot \bar{d} : \alpha \geq 0\}$ holds for some $i \in I$, then $\bar{x}$ is on a line segment between x^i and $x^i + \bar{\alpha} \cdot \bar{d}$ for $\bar{\alpha} > 0$ sufficiently large, which contradicts that $\bar{x}$ is an extreme point of K .

Hence $\bar{x} \notin \{x^i + \alpha \cdot \bar{d} : \alpha \geq 0\}$ for all $i \in I$. In particular $\bar{x} \neq x^i + \bar{d}$ for all $i \in I$, and therefore (1) gives $\bar{x}$ as a convex combination of the points $x^i + \bar{d} \in K \setminus \{\bar{x}\}$ for $i \in I$ again contradicting that $\bar{x}$ is extreme for K . ■

3 Valid inequalities for the cone extension

There is also a close relationship between the valid inequalities for a convex set and the valid inequalities for its cone extension. This relationship is described in the following theorem.

Theorem 3 (*Valid inequalities for K^{ext} vs. the valid inequalities for K*)

Let $\alpha^T x \leq \beta$ be an arbitrary inequality.

(1) If K is non-empty and $\alpha^T x \leq 0$ is not valid for $\text{rec}(K)$:

$\alpha^T x \leq \beta$ is not valid for K .

(2) If $\alpha^T x \leq 0$ is valid for $\text{rec}(K)$:

$\alpha^T x \leq \beta$ is valid for $K \iff \alpha^T x - \beta \cdot \lambda \leq 0$ is valid for K^{ext} . ■

Proof: (1): If K is non-empty and $\alpha^T d > 0$ holds for some $d \in \text{rec}(K)$, then one has $\{x + t \cdot d : t \geq 0\} \subseteq K$ and $\lim_{t \rightarrow +\infty} \alpha^T(x + t \cdot d) = +\infty$ for all $x \in K$. Therefore, if $\alpha^T d > 0$ for some $d \in \text{rec}(K)$, then $\alpha^T x$ is not bounded above on K , and this excludes validity of any inequality $\alpha^T x \leq \beta$ for K that has α as normal vector.

(2): " $\Leftarrow$ ": Suppose $\alpha^T x - \beta \cdot \lambda \leq 0$ is valid for K^{ext} . This implies $\alpha^T x - \beta \leq 0$ for all $\begin{pmatrix} x \\ 1 \end{pmatrix} \in K^{\text{ext}}$. Since $K = \{x \in \mathbb{R}^n : \begin{pmatrix} x \\ 1 \end{pmatrix} \in K^{\text{ext}}\}$, this implies $\alpha^T x \leq \beta$ is valid for K .

" $\Rightarrow$ ": Now suppose $\alpha^T x \leq \beta$ is valid for K . We must show $\alpha^T x - \beta \cdot \lambda \leq 0$ is valid for K^{ext} . Therefore let $\begin{pmatrix} y \\ \mu \end{pmatrix} \in K^{\text{ext}}$ be arbitrary. We must show $\alpha^T y - \beta \cdot \mu \leq 0$. There are two cases:

(a) " $\mu = 0$ ": Hence $\begin{pmatrix} y \\ \mu \end{pmatrix} = \begin{pmatrix} y \\ 0 \end{pmatrix} \in K^{\text{ext}}$, and $y \in \text{rec}(K)$ by definition of K^{ext} .

Since $\alpha^T x \leq 0$ is valid for $\text{rec}(K)$, one obtains $\alpha^T y - \beta \cdot \mu = \alpha^T y \leq 0$.

(b) " $\mu > 0$ ": This implies $\begin{pmatrix} y \\ \mu \end{pmatrix} = \mu \cdot \begin{pmatrix} y/\mu \\ 1 \end{pmatrix}$. Hence $\begin{pmatrix} y/\mu \\ 1 \end{pmatrix} \in K^{\text{ext}}$, so Definition 1 gives $\frac{y}{\mu} \in K$, and therefore $\alpha^T \frac{y}{\mu} \leq \beta$ follows from validity of $\alpha^T x \leq \beta$ for K . We conclude that $\alpha^T y - \beta \cdot \mu \leq 0$ holds. ■